SET A

Department of Computer Science and Engineering
CS2006: Operating System (OS)
Quiz II – Answer Sheet

Roll No: __________ **Name:** __________ **Batch:** __________

Time: 40 minutes **Maximum Marks: 30**

Note: Each subjective question carries 2 marks while an MCQ question carries 1 mark.

1. Consider a demand paging system with three page frames (initially empty) and LRU page replacement policy. For the following page reference string:
   1, 2, 3, 4, 2, 1, 5, 3, 2, 4, 6
   What is the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place)?
   Ans: Page fault = 10; total access = 11; page fault rate = 10/11 = 0.91 or 0.9

2. Consider a system with a two-level paging scheme in which a regular memory access takes 120 nanoseconds, and servicing a page fault takes 6 milliseconds. An average instruction takes 80 nanoseconds of CPU time and requires two memory accesses. The TLB hit ratio is 92%, and the page fault rate is 1 in every 20,000 instructions. What is the effective average instruction execution time?
   Ans: 658.4 or between 658.0 to 658.9

3. Consider a 64-bit machine where a three-level paging scheme is used. If the hit ratio to the TLB is 95%, and it takes 10 nanoseconds to search the TLB and 120 nanoseconds to access the main memory, what is the effective memory access time (EMAT) in nanoseconds?
   Ans: 148

4. A computer system implements 4 kilobyte pages and a 64-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 32 megabytes, the length of the virtual address supported by the system is __________ bits.
   Ans: 34

5. Consider a system with byte-addressable memory, 64-bit logical addresses, 8 kilobyte page size, and page table entries of 8 bytes each. The size of the page table in the system in gigabytes is __________.
   Ans: 2^{24} GB = 16 777 216 GB

6. In a system with segmented paging, the logical address is divided into:
   (A) Segment number, Offset
   (B) Segment number, Page number, Offset
   (C) Page number, Frame number
   (D) Page number, Offset

7. A system uses paging with 32-bit logical address space and 4 KB page size. How many entries will be there in the page table for a process?

(A) 2^{10}
(B) 2^{12}
(C) 2^{20}
(D) 2^{32}

8. Which of the following memory management schemes suffers from both internal and external fragmentation?
(A) Paging
(B) Segmentation
(C) Contiguous Allocation
(D) Paging with variable page size

9. Consider a system with demand paging. If the page fault service time is 8 ms and memory access time is 200 ns, what should be the maximum page fault rate to keep the effective memory access time less than 220 ns?
(A) 0.001
(B) 0.00025
(C) 0.00002 – (Nearest Answer)
(D) 0.00005
Ans:

$$\text{EAT} = (1-p)\,200\text{ ns} + p\,(8\text{ ms} + 200\text{ ns}) < 220\text{ ns}$$

$$200 + p \cdot 8\,000\,200 < 220 \implies p < \frac{20}{8\,000\,200} \approx 2.5 \times 10^{-6} \text{or } 0.00\,00\,025$$

$$\boxed{\text{No given option meets } p < 2.5 \times 10^{-6} \text{or } 0.00\,00\,025}$$

Note: Since none of the provided options is correct, grace marks will be awarded to all who attempted the question.

10. Which memory allocation strategy allocates the largest available hole in memory to a process?
(A) First Fit
(B) Best Fit
(C) Worst Fit
(D) Next Fit

11. A disk has 12 surfaces, 300 tracks per surface, 512 sectors per track, and 1024 bytes per sector. If the format overhead per sector is 64 bytes, then the formatted capacity of the disk is __________ bytes.
Ans: Raw Capacity = 12 × 300 × 512 × 1024 = 188 74 36 800 bytes;
Formatted Capacity = 12 × 300 × 512 × (1024 - 64) = 176 94 72 000 bytes

12. A disk has an average seek time of 5 ms, rotates at 7200 RPM, and has 512 bytes per sector, where each track contains 2^{12} sectors. If a file of 2 MB is stored contiguously, the time taken to read the file is __________ milliseconds.
Ans: 17.5 ms

13. A disk has an average seek time of 10 ms and rotates at 5400 RPM. If a read operation requires seeking + 1 full rotation + data transfer time of 8 ms, then the total time to complete the read operation is __________ milliseconds.
Ans: 10 + 11.11 + 8 = 29.11 ms

14. Let the diameter of the innermost track of a disk is 21 cm. If the recording density of the track is 2 KB/cm, then the track capacity of the innermost track is __________ KB.
Ans: Track Capacity $= 2 \times 3.1416 \times 10.5 \times 2 = 131.9472$ KB

15. If a disk rotates at 5400 RPM and has 512 sectors per track with a sector size of 512 bytes, then the data transfer rate is __________ MB/s.
Ans: Given; 5400 RPM
$\Rightarrow$ 5400 rotations in 1 minute (60 sec)
$\Rightarrow$ 1 rotation in $\left(\frac{60}{5400}\right)$ seconds
So, Transfer rate $= \frac{256 \text{ KB}}{60/5400 \text{ sec}} = 23040$ KB/sec
$\Rightarrow \frac{23040}{1024} = 22.5$ MB/sec

16. A certain hard drive rotates at 6000 rpm. It has 1KB per sector and average 128 sectors per track. Which of the following statements is/are true?
I) The average latency of the drive is under 8 milliseconds
II) The burst data rate of the drive is over 10MM/sec
III) The average capacity per track is over 1MB

(A) II only
(B) I and II only
(C) II and III only
(D) I, II and III

17. Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each section holds 1024 bytes of data. The number of cylinders in hard disk is __________
(A) 4096
(B) 2048
(C) 1024
(D) 512

18. The question given below concern a disk with a sector size of 512 bytes, 2000 tracks per surface, 50 sectors per track, five double-sided platters, and average seek time of 10 milliseconds.
Given below are two statements:
Statement I: The disk has a total number of 2000 cylinders.
Statement II: 51200 bytes is not a valid block size for the disk
In the light of the above statements, choose the correct answer from the options given below:
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I correct and Statement II is false
(D) Statement I incorrect and Statement II is true

19. Which factor does NOT directly affect disk access time?
    (A) Seek time
    (B) Rotational latency
    (C) Data transfer rate
    (D) Monitor refresh rate

20. If the disk rotates at 7200 RPM, what is the average rotational latency?
    (A) 8.33 ms
    (B) 4.17 ms
    (C) 1 ms
    (D) 0.5 ms